NO.18 COOLING SYSTEM CONCERNS-RUNS COLD [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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18	COOLING SYSTEM CONCERNS-RUNS COLD
DESCRIPTION	• Engine does not reach normal operating temperature.
POSSIBLE CAUSE	• Improper indication of low engine coolant temperature indicator light and/or high engine coolant temperature warning light — ECT sensor malfunction — Instrument cluster malfunction • Cooling fan system malfunction • Condenser fan system malfunction • Thermostat malfunction (stuck at fully open) • Foreign material penetration in engine coolant

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**DETERMINE IF MALFUNCTION CAUSE IS HVAC SYSTEM OR ENGINE** • Is a customer complaint “Lack of passenger compartment heat” only?	Yes	Inspect the A/C heater system. Repair or replace the malfunctioning part according to the inspection results.
		No	Go to the next step.
2	**VERIFY IDLE SPEED** • Does the engine speed continue at fast idle?	Yes	Perform the symptom troubleshooting “NO.9 FAST IDLE/RUNS ON”. (See NO.9 FAST IDLE/RUNS ON [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
3	**INSPECT ECT SENSOR SIGNAL** • Access the ECT PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Compare the ECT PID and low engine coolant temperature indicator light and high engine coolant temperature warning light operation. • Is the ECT PID value same as indicator/warning light readings?	Yes	Go to the next step.
		No	Inspect the ECT sensor. (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If there is any malfunction: — Replace the ECT sensor. (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If there is no malfunction: — Perform the instrument cluster symptom troubleshooting “LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT/HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT ILLUMINATES OR FLASHES CONTINUOUSLY”. (See LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT/HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT ILLUMINATES OR FLASHES CONTINUOUSLY [INSTRUMENT CLUSTER].)
4	**INSPECT COOLING FAN CONTROL SYSTEM OPERATION** • Perform the Cooling Fan Control System Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Does the cooling fan control system operate properly?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results.

STEP	INSPECTION	RESULTS	ACTION
5	**INSPECT THERMOSTAT** • Cool down the engine. • Remove the thermostat. (See THERMOSTAT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Inspect the thermostat operation. (See THERMOSTAT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the thermostat. (See THERMOSTAT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Reinstall the thermostat, then go to the next step. (See THERMOSTAT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
6	**VERIFY THAT NO FOREIGN MATTER IS IN ENGINE COOLANT** • Drain the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Has foreign matter penetrated the engine coolant?	Yes	Replace the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
7	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)		